



UNIVERSITY OF SRI JAYEWARDENEPURA
FACULTY OF APPLIED SCIENCES

Bachelor of Science Degree First Year First Semester Course Unit Examination –
September 2025

DEPARTMENT OF COMPUTER SCIENCE

ICT 1011 Computer Programming

Time: One (01) hour	No. of questions: 03	No. of pages: 03	Total marks: 100
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Instructions:

1. Answer ALL questions
 2. Candidates should start a new question on a new page.
 3. Clearly mention the question number, part number and sub part number [E.g. (Q1) (a), (Q1)(a)(i)].
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Question 01 (25 marks)

- I. Identify and describe four (04) significant characteristics of pseudocode. (04 marks)
- II. Write pseudocode solutions for the following problems:
 - (a) Determine the largest number among three given numbers. (07 marks)
 - (b) Generate and display the multiplication table of a given number. (04 marks)
 - (c) Check whether a given number is a prime number. (*Note: A prime number is divided only by 1 and by itself without leaving a remainder.*) (10 marks)

Question 02 (40 marks)

- I. Write the output of the following code segments. If it leads to a blank output, clearly mention the reason for it.

(a)

```
public class Test1 {
    public static void main(String[] args) {
        int a = 7;
        int b = 2;
        double result = a / b;
        System.out.println(result);
    }
}
```

(02 marks)

(b)

```
public class Test2 {  
    public static void main(String[] args) {  
        String s1 = "Hello";  
        String s2 = s1;  
        s1 = s1.concat(" World");  
        System.out.println(s1);  
        System.out.println(s2);  
    }  
}
```

(03 marks)

(c)

```
public class Test3 {  
    public static void main(String[] args) {  
        int x = 5;  
        int y = x++ + ++x;  
        System.out.println("x = " + x + ", y = " + y);  
    }  
}
```

(04 marks)

(d)

```
public class Test4 {  
    public static void main(String[] args) {  
        int[] arr1 = {1, 2, 3};  
        int[] arr2 = arr1;  
        arr2[1] = 10;  
        System.out.println(arr1[1]);  
    }  
}
```

(02 marks)

(e)

```
public class Test5 {  
    public static void main(String[] args) {  
        String s2 = "Java";  
        String s3 = "Java";  
        System.out.println(s2 == s3);  
        System.out.println(s2.equals(s3));  
    }  
}
```

(04 marks)

II. You are tasked to develop Smart Parking Fee Calculator using Java programming language. The city parking lot charges customers according to the following rules:

- Grace period: The first 15 minutes of parking are free.
- Slot-based charging: After the grace period, the fee is Rs. 120 per 30 minutes or part thereof.
- Daily cap: The maximum fee per calendar day is Rs. 1,500.
- Stay cap: The total fee across all entered days must not exceed Rs. 3,500.

Write a Java program to display the output in Figure 1 by doing the following (Implement (b), (c), and (e) inside main method.):

- (a) Define a class with the parking charge parameters (grace period, slot rate, daily cap, overall cap) and include the main method to implement the other tasks. (05 marks)

- (b) Read the number of days the vehicle has been parked within a week. (03 marks)
- (c) For each day, read the number of minutes the vehicle was parked. (03 marks)
- (d) Write a function that accepts the total parking time in minutes as a parameter, calculates the daily parking fee, and displays the result. (Hint: For this calculation, you may use the Java method `Math.ceil(x)`, which always rounds a decimal number upwards to the nearest integer to ensure that any fraction of a 30-minute slot is charged as a full slot. Example: `Math.ceil(2.1) = 3`). (09 marks)
- (e) Calculate and display the total parking fee for the entire stay. (05 marks)

```

Enter number of days parked (within a week): 3
Enter minutes parked on day 1: 5
Enter minutes parked on day 2: 20
Enter minutes parked on day 3: 80

Day 1 fee: Rs. 0
Day 2 fee: Rs. 120
Day 3 fee: Rs. 360
Total parking fee for the stay: Rs. 480

```

Figure 1: Output of the Smart Parking Fee Calculator

Question 03 (35 marks)

- I. Briefly explain two (02) different approaches exist in Java to store data. (02 marks)
- II. A company records the daily production (in units) of a factory for 4 weeks, and each week has 7 days. Write a Java program to display the output in Figure 2 by doing the following:
 - (a) Accept and store the data. (14 marks)
 - (b) Display the total production of each week. (08 marks)
 - (c) Find and display the week with the highest total production. (05 marks)
 - (d) Find and display the lowest daily production value across all 4 weeks and the day/week it occurred. (06 marks)

```

Enter production values for 4 weeks (7 days each):
Week 1: 120 130 140 200 180 160 170
Week 2: 110 115 120 130 140 150 160
Week 3: 200 210 190 180 175 165 155
Week 4: 100 90 95 85 80 75 70

Total of Week 1: 1100
Total of Week 2: 925
Total of Week 3: 1275
Total of Week 4: 595
Week with highest total production: Week 3 (1275)
Lowest daily production: 70 (Week 4, Day 7)

```

Figure 2: Output of the Java program

*****END*****